

Deep level transient spectroscopy and theoretical modelling of defect states in few-layer MoS₂

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ABSTRACT

Native defects can essentially affect the properties of semiconductors and devices based on them. The defect influence is critical for 2D materials obtained by mechanical exfoliation from layered crystals, as most defects may be introduced when exfoliating. A film of few-layer MoS₂ flakes on a SiO₂/Si substrate was studied using deep-level transient spectroscopy (DLTS). A set of electron traps with energy levels at 303, 440, and 633 meV below the conduction band was found. The values are compared to those obtained by the density functional theory calculations of most abundant point defects in bilayer MoS₂, such as Mo and S vacancies, Mo+S and S divacancies, or O substituting S in a surface S layer. Based on the calculation results, the three states found by DLTS were attributed to S vacancy (440 meV) and S divacancy (303 and 633 meV), being the most expected when preparing the layered 2D structures by mechanical exfoliation.

1. Introduction

Since the discovery of graphene, the whole universe of 2D materials has attracted immense attention. As the former has rather narrow or zero bandgap, hindering its use in nano-optoelectronics, other promising materials possessing a wider bandgap are of great interest. Due to the development of exfoliation techniques, there is a feasibility to obtain 2D materials from layered crystals, such as transition metal dichalcogenides (TMDCs), the intralayer binding energy of which is much higher than the interlayer one [1–3]. With indirect bandgap in bulk, they become direct-bandgap materials being thinned to monolayer (1L) sheets.

Molybdenum disulfide (MoS₂) is now one of the most studied 2D materials [4]. In the 1L case it has a fairly wide bandgap around 1.9 eV and can emit light in the visible range; thus, it is applicable as an active element in nano-optoelectronic devices [5–9]. Even though only 1L MoS₂ acquires a direct bandgap and is thus most intensively investigated

[10,11], few-layer (FL) flakes exhibit satisfactory luminescent properties [12–15] and are also widely studied [16,17]. Moreover, the efficiency of thin 2D MoS₂ layers as light emitters and detectors can be enhanced in contact with metal [18–21] or semiconductor nanoparticles and quantum dots [22–27]. In addition, the photoelectric properties of 2D MoS₂ can be tuned by other ways of doping, e.g. excitation power [15,17,28,29] and energy [15,26,30], as well as heating [11,14,17,28] or mechanical strain. The effect of strain on doping and, hence, on the optical properties is well studied [30–35], since atomically-thin MoS₂ is considered as a promising material for flexible electronics.

It is known that 2D MoS₂ is intrinsically an *n*-type semiconductor due to inherent defects [36,37]. These defects also attract special attention, as they may essentially affect the properties and sometimes worsen the performance of MoS₂-based devices. The point defects and their complexes in 2D MoS₂ were observed by high-resolution electron microscopy [37–41], scanning tunneling microscopy [42–45], or X-ray photoelectron spectroscopy [46–49] and widely studied theoretically

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[38–40,50–52], when calculating such demanded parameters as formation energy and energy levels introduced into the band structure. Nevertheless, the experimental studies of the defect energy levels in 1L and FL MoS₂ by transient spectroscopies are rare [41] due to obvious difficulties in preparing the nanometer-thick samples with few-micrometer lateral sizes for the electrical measurements. Most of the recent deep level transition spectroscopy (DLTS) studies of MoS₂ were applied to bulk MoS₂ or at least multilayers of tens or hundreds of nanometer thickness [40,53].

In this work, we used DLTS to identify defect states in FL MoS₂ flakes, which were exfoliated from high-quality bulk crystals and deposited from the solution onto a SiO₂/Si substrate to form a thin film over few square millimeters. Electron traps with energy levels at 303, 440, and 633 meV below the conduction band edge (E_c) were found. Using the results of density functional theory (DFT) calculations, the traps were identified as S vacancy (V_S) and S divacancy (V_{S2}).

2. Methods

2.1. Sample preparation

To fabricate the Ag/MoS₂/SiO₂/p-Si structure for DLTS measurements, we used a commercially available (Graphene Laboratories Inc., USA) ethanol solution of MoS₂ flakes (18 mg/L) prepared using solution-based exfoliation. The range of flake thicknesses declared by the manufacturer was from 1 to 8 layers, and the lateral size varied from 100 to 400 nm (Fig. S1, Supplementary Materials). The positions of the A and B exciton peaks in the absorption spectra (Fig. S2) correspond to those characteristic for FL MoS₂ [54]. The flakes were deposited by spray-coating onto a p-Si (7.5 $\Omega\cdot\text{cm}$) substrate at 80 °C. The substrate was pre-cleaned by immersion into 48 % HF acid, forming a thin natural silicon oxide layer at the substrate surface during MoS₂ deposition in ambient air. The Raman spectrum of deposited MoS₂ shows the E_{2g}^1 and A_{1g} Raman modes (Fig. S3), while the relatively wide frequency difference of $\sim 24\text{ cm}^{-1}$ between them may evidence conglomeration of the flakes.

2.2. Deep level transition spectroscopy

To study the electrical properties of MoS₂/SiO₂/Si heterostructures, a 100-nm thick Ag circle electrode with area of $\sim 7\text{ mm}^2$ was deposited onto of the MoS₂ thin film in vacuum through a shadow mask with metal silver (99.99 % purity) pellets by thermal evaporation. The deposition pressure was $2.0\cdot 10^{-5}$ Torr. The deposition rate (about 1 nm/s) was measured with a quartz crystal microbalance. An In/Ga eutectic back electrode was fabricated onto the back Si surface of the SiO₂/Si substrate at room temperature. The schematic of the obtained diode-like structure is displayed in Fig. 1a. An Ag/SiO₂/Si structure without MoS₂ was studied as a reference sample. The C-V and J-V characteristics of the device at different temperatures were investigated using an Agilent 4284A LCR meter and an Agilent 4156C Semiconductor Parameter Analyzer, respectively.

The energy spectrum of the defect states was studied using DLTS by a digital DLTS spectrometer FT-1030. Transient capacitance signals were detected with a 1 MHz capacitance meter (Boonton 72B C-meter). The DLTS spectra were obtained from Fourier transforms of the capacitance transients recorded as a function of temperature in the range of 80–320 K. This technique [55] is based on numerical Fourier transformation of the measured capacitance transients originated from the temperature-dependent kinetics of charge carrier release from in-gap states after filling them by electric pulses (Fig. 1b). The emission time constants were calculated from the maximum peaks of the Fourier coefficients versus temperature. After identifying all the peaks, an Arrhenius plot of $\ln(\tau_{\text{peak}} T^2)$ versus $1/kT$ yields the trap energies from the slope of the linear fit. The defect levels were filled by a 100 μs wide voltage pulse, $V_B = +0.5\text{ V}$, while the diode was under a reverse bias of $V_B = -1.0\text{ V}$ before and after the pulse. The Schottky barrier at the interface near the top Ag contact, as well as the filling and release of the traps in the MoS₂ space charge area, are under positive and negative bias, respectively, as shown in Fig. 1b, where the band bending at the interfaces is schematically illustrated considering work functions of 5.2–5.4 eV for FL MoS₂ [56], 4.3 eV for Ag [57], 4.6–4.9 eV for Si [58], p-doping of the SiO₂/Si substrate and expected n-doping of FL MoS₂ and their bandgap widths of 1.1 and 1.4–1.6 eV [10], respectively.

3. Results and discussion

Capacitance-voltage (C-V) and current-voltage (J-V) characteristics were obtained at room temperature to evaluate the potential barrier in the formed Ag/MoS₂/SiO₂/Si heterostructure. The C-V and J-V plots of the Ag/MoS₂/SiO₂/Si and Ag/SiO₂/Si metal-insulator-semiconductor (MIS) diodes at 300 K are shown in Figs. 2a and 2b, respectively. The J-V curves are expectably nonlinear, i.e. the current under the forward bias (a negative potential to the MoS₂ film) was essentially higher than the current under the reverse bias. The rectifying behavior was observed for all studied temperatures indicating to the presence of the depletion layer in the silicon substrate. Furthermore, the reverse current of the MoS₂/SiO₂/Si MIS structure was one order less compared to that of the reference structure without the MoS₂ layer, obviously due to a higher potential barrier caused by a greater work function of MoS₂ compared to Si (a band schematic at Ag interface to SiO₂/Si is given in Figure S4 of Supplementary Materials).

We analyzed the C-V curve shapes measured using a probe signal frequency of 1 MHz to estimate the height of the potential barrier. The plots of $1/C^2$ versus voltage in Fig. 2a contain linear sections under the reverse bias (the deviations from the linearity are due to insufficient depletion at low voltages leading to a lower capacitance). Their intersections with the voltage axis indicate potential barrier heights of 109 and 373 meV for the Ag/SiO₂/Si and Ag/MoS₂/SiO₂/Si diodes, respectively. Consequently, the deposition of a MoS₂ film onto a p-type Si substrate causes a higher potential barrier than that in the reference diode.

DLTS spectra (the first order Fourier sine coefficient b_1 vs temperature) of the Ag/MoS₂/SiO₂/Si diode at different rate windows of 50 and

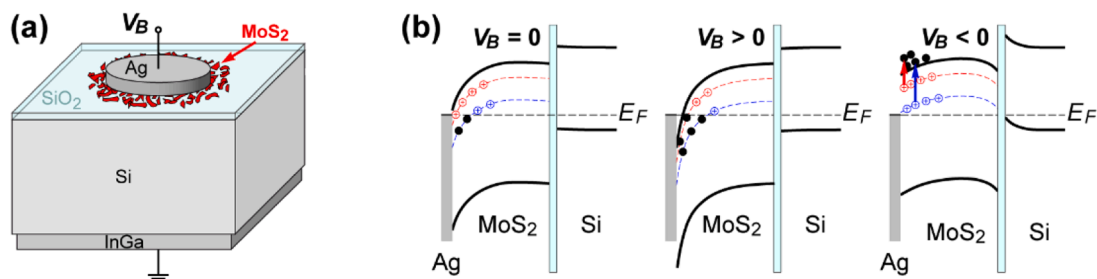


Fig. 1. (a) Diode-like structure with MoS₂ flakes for electrical and DLTS experiments; (b) band bending at MoS₂/Si interface with two types of traps at different biases (the carrier release contributing to the DLTS response is shown by arrows).

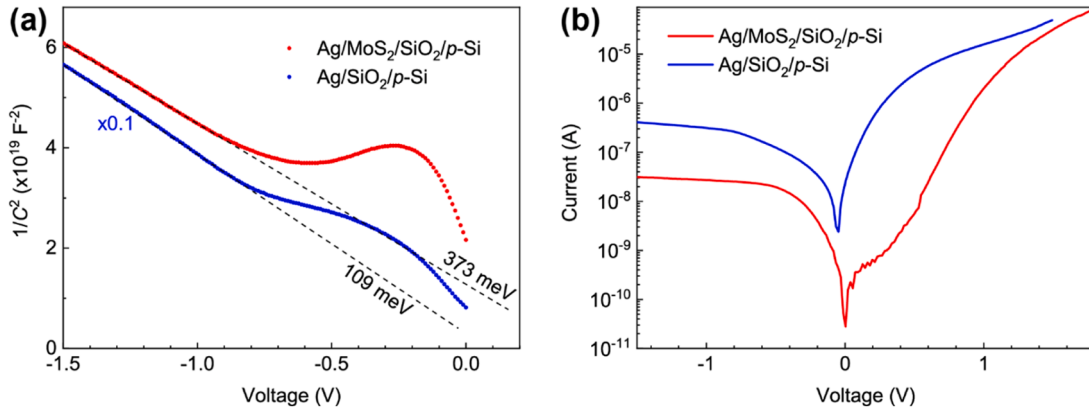


Fig. 2. C-V (a) and J-V (b) curves of the Ag/MoS₂/SiO₂/Si and Ag/SiO₂/Si MIS diodes at 300 K.

300 ms are shown in Fig. 3a. The dependences include three peaks, meaning that at least three trap states (energy levels) contribute to the spectra. Their positions are estimated from the slopes of the plots $\ln(\tau_{\text{peak}} T^2)$ versus $1/kT$ for different rate times, where τ_{peak} is the relaxation time at the temperature of the respective peak in the DLTS spectrum. Since the positive filling pulse ($U_F = +0.5 \text{ V}$) fills the MoS₂ states by electrons, and the capacitance increases (positive transient signal), we may conclude that after injection electrons are released and traps are emptied. Therefore, the observed DLTS peaks are associated with electron traps. The plots for all three components shown in Fig. 3b are linear and give the positions of the electron levels in the MoS₂ film at $E_c - 303 \pm 5$, 440 ± 5 , and $633 \pm 5 \text{ meV}$. It is worth noting that the DLTS spectrum of the reference Ag/SiO₂/Si structure in Fig. 4a contains only one peak corresponding to a state positioned at $E_c - 377 \pm 5 \text{ meV}$ as follows from the Arrhenius plot in Fig. 4b. Studying the Ag/MoS₂/SiO₂/Si structure, this peak is obviously lost against the higher response of MoS₂ observed in Fig. 3a at the same temperatures.

As for identifying our electron traps, different kinds of V_S are the most probable origin. Indeed, they are most expected in MoS₂ flakes exfoliated from a high-quality bulk crystal, as each MoS₂ monolayer consists of an internal Mo layer and two external S layers, thus, the surface S layers are most possibly damaged when mechanically or chemically exfoliated. Furthermore, the formation energy of V_S is much lower than those of other point defects, except for S adatoms [39,50]. Estimations of the defect concentrations confirm these considerations. V_S are the most numerous defects not only in exfoliated flakes but also in CVD-grown ones [39]. This is a possible reason why MoS₂ reveals mainly electron conductivity without intentional doping [36,37].

To identify the defects that created the levels found in the

experiment, DFT modeling of the energy band structure in a bilayer MoS₂ with different expectable defects was performed (Methods, Supplementary Materials). Apart from V_S and related complexes, the levels of V_{Mo} and of oxygen (O) substituting S (O_S) in an external layer were calculated. The calculated band structures are given in Fig. 5a and the respective densities of states (DOS) in Fig. 5b. The O_S defect is not found to introduce any states in the bandgap, while the levels introduced by V_{Mo} should be acceptors, thus, only V_S is of interest. It should be added that no contribution of Ag atoms to the DLTS response was considered, as Ag doping of MoS₂ was found to cause a p -doping of this material [59], while the DLTS signal polarity implies just donor-like responsible levels.

The states of the most expectable V_S in an external S layer are at $E_c - 430/490 \text{ meV}$, while DOS shows a higher population of the deepest level. The location of V_S on an internal or external S layer affects the level position insignificantly, the states in an internal layer are estimated to be at $450/530 \text{ meV}$ (its DOS is presented in Fig. 5b); however, such a defect is much less probable. A complex of two neighbor S vacancies (V_{S2} divacancy) in an S layer introduces two groups of states at $E_c - 156/324$ and $\sim 600 \text{ meV}$; moreover, the DOS points out that these levels are populated. So, based on our calculations, we may attribute the found state $E_c - 440 \text{ meV}$, which has the highest contribution to the DLTS signal, in FL MoS₂ to V_S in an external S layer. As for the two other states, at $E_c - 303$ and 633 meV , revealing lower responses in the DLTS spectra, we relate them to V_{S2} , which has an order lower content in 2D MoS₂, according to the estimations by Zhao et al. [41].

It is worth noting that other researchers also unquestionably related the experimentally found states at $270\text{--}360 \text{ meV}$ to V_S [36,40,53], while Zhao et al. [41] attributed a state found at 632 meV also to V_S . Otherwise, we should note that, apart from the level at 270 meV related to V_S ,

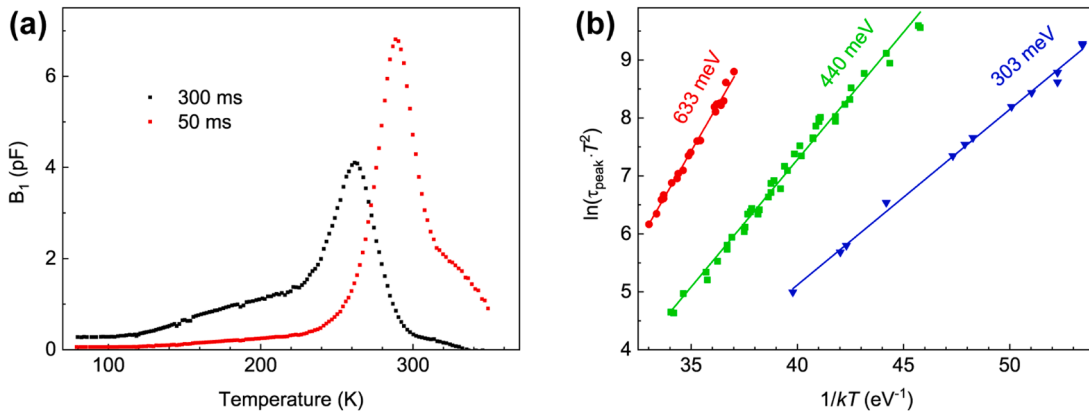


Fig. 3. (a) DLTS spectra of the Ag/MoS₂/SiO₂/Si diode obtained under filling pulse with an amplitude of $U_F = +0.5 \text{ V}$ and a width of $100 \mu\text{s}$ at different transient periods. (b) Arrhenius plot of the deep traps found in the spectra.

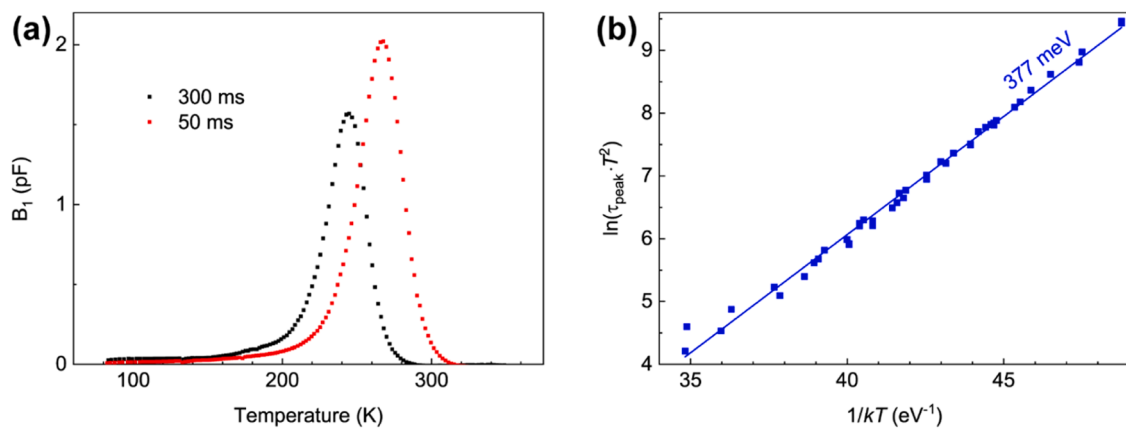


Fig. 4. (a) DLTS spectra of the reference Ag/SiO₂/Si diode obtained under filling pulse with an amplitude of $U_F = +0.5$ V and a width of 100 μ s at different transient periods. (b) Arrhenius plot of the found state.

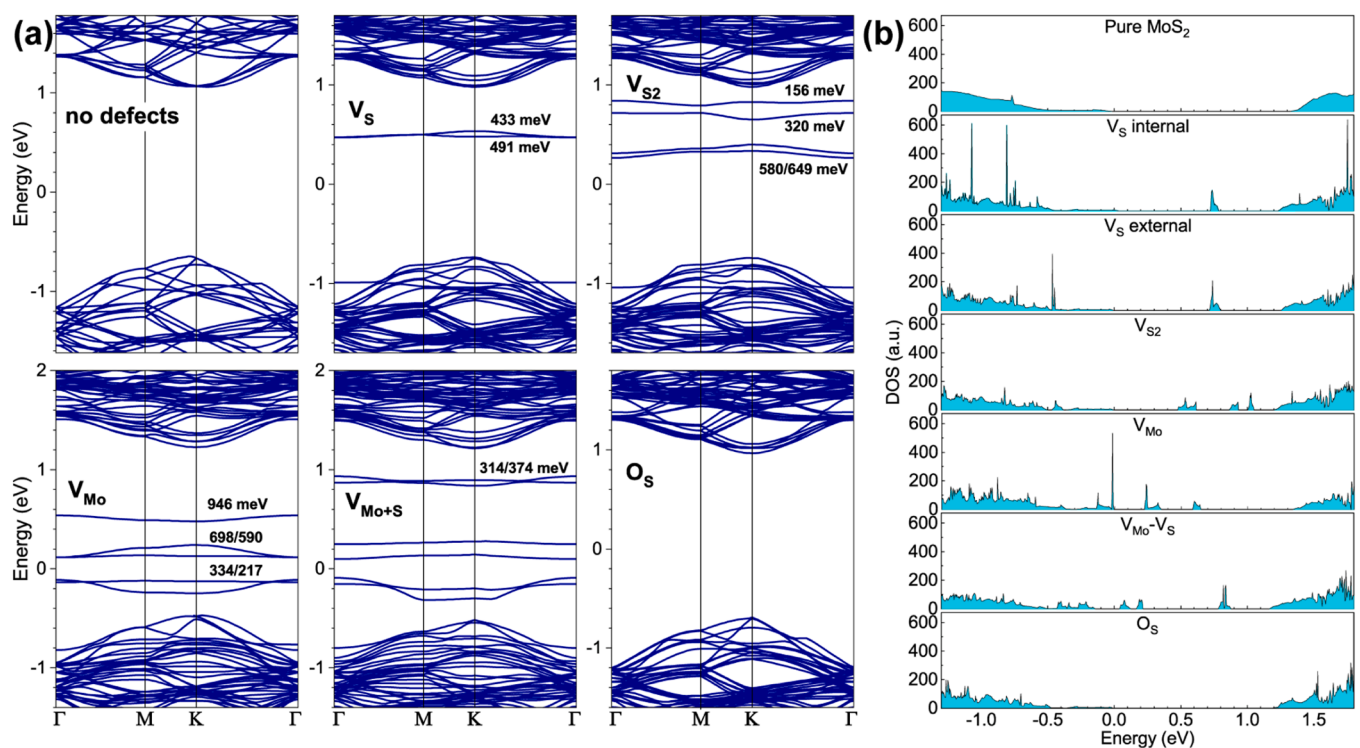


Fig. 5. DFT-calculated (a) band structure and (b) DOS of a perfect bilayer MoS₂ compared to that with different types of defects, the level depths below the conduction band near the K-point of the Brillouin zone are indicated nearby.

Ci et al. [40] detected a state at 440 meV and attributed it to a DX center, i.e. a complex consisting of a donor-like center and a vacancy. However, this state revealed that the DLTS response was several times less than that of V_S found there. Therefore, we cannot similarly identify our state with the same position at $E_c - 440$ meV.

So, the studied 2D-MoS₂ flakes demonstrate abundance of V_S , while no other defects are found with the DLTS technique. It should be emphasized that the detected DLTS signal from MoS₂ can be essentially lowered due to the natural oxidation of MoS₂ after exfoliation. Although MoS₂ crystal surface is stable to oxidation due to large energy barrier [60], V_S are easy passivated by oxygen, including atmospherical one [60,61]. As O_S defects introduce levels into the valence band rather than the bandgap [62] and induce p -conductivity in MoS₂ [63], the oxidation is considered as a promising way for elimination of the negative effect of V_S [62–65].

4. Conclusions

Defects in thin films of exfoliated FL MoS₂ flakes drop-casted onto a Si substrate were studied using the DLTS technique. Three electron traps with deep energy levels at $E_c - 303$, 440, and 633 eV were detected just in the film, while the pristine substrate revealed a different value of the activation energy. To identify the found level depths with native MoS₂ defects, DFT calculations of the most abundant point defects in bilayer MoS₂ were performed. Comparing to the calculated data, the energy states found by DLTS were attributed to S vacancy (440 eV) and S divacancy (303 and 633 eV), being most expected when preparing the layered 2D crystalline structures with S atoms at the external sides.

CRedit authorship contribution statement

Serhiy Kondratenko: Conceptualization, Resources, Methodology,

Investigation, Formal analysis, Writing – original draft, Supervision, Project administration, Funding acquisition. **Oleksandr I. Datsenko**: Formal analysis, Writing – original draft, Visualization. **Sergii Golovynskyi**: Formal analysis, Writing – original draft, Visualization. **Anastasiya Mykytiuk**: Investigation, Visualization, Writing – review & editing. **Artem Kuklin**: Formal analysis, Visualization, Writing – review & editing. **Hans Ågren**: Resources, Writing – review & editing. **Voldymyr Dzhagan**: Resources, Writing – review & editing. **Dietrich R.T. Zahn**: Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.surfin.2025.106928](https://doi.org/10.1016/j.surfin.2025.106928).

Data availability

Data will be made available on request.

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